NexusOS

AI Business Operating System

Part 03.0

Linear Issue WAL-7

Database Schema & Enterprise Data Model

Project	NexusOS — AI Business Operating System
Part	Part 03.0
Linear issue	WAL-7
Report title	Database Schema & Enterprise Data Model
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Source report(s)	docs/reports/Part-03.0/Part-03.0-Master-Engineering-Summary.md

This PDF is a presentation export of repository engineering report content. Source report files in the repository were not modified.

Evidence Statement

This report is a retrospective engineering reconstruction produced after implementation. Its contents are derived exclusively from repository evidence. It is not a contemporaneous engineering document. No engineering events are fabricated. Where repository evidence is unavailable, that limitation is explicitly documented.

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Report Content

Content below is copied from the source repository report(s) without alteration.

Part 03.0 — Master Engineering Summary

Linear: WAL-7 (umbrella) **Milestone:** Part 03 — Authorization & Multi-Tenant Data Platform **Reconstruction date:** 2026-07-19

Evidence Statement

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1. Purpose of Part 03.0

Deliver the enterprise multi-tenant PostgreSQL schema, RBAC catalog, repository tenant-isolation foundation, Alembic baseline, and product documentation that NexusOS authentication (Part 02) and security hardening (Part 03.1) build on.

2. What the repository contains

Deliverable	Evidence
37-table initial schema	Alembic revision `0a961c2ea2cf` — 37 `op.create_table` calls; 150 `op.create_index` calls
SQLAlchemy models	`backend/app/models/*` registered in `models/__init__.py`
Shared standards	`TenantBusinessMixin` / naming conventions in `backend/app/models/base.py`
UUID v7	`backend/app/utils/uuid7.py`
Tenant repository foundation	`backend/app/repositories/base.py` (`_base_query` fail-closed)
Permission seed	`backend/app/db/seed.py` — **53** permissions; roles Owner/Admin/Manager/Employee/Viewer
Isolation tests (original at landing)	5 tests in `838db64` version of `tests/test_tenant_isolation.py`
Product documentation	Originally `docs/database.md` in `838db64`; now `docs/product/database.md` (docs reorg `db3cb86`)
Primary git landing	`838db64` (2026-07-09)

3. Phase map (Linear ↔ reports)

Linear	Phase	Report
WAL-8	Requirements Analysis	`Part-03.0-Requirements-Analysis.md`
WAL-9	Architecture Design	`Part-03.0-Architecture-Design.md`

WAL-10	Database Modeling	`Part-03.0-Database-Modeling.md`
WAL-11	Schema Implementation	`Part-03.0-Schema-Implementation.md`
WAL-12	Migration Development	`Part-03.0-Migration-Development.md`
WAL-13	Validation & Testing	`Part-03.0-Validation-and-Testing.md`
WAL-14	Engineering Review	`Part-03.0-Engineering-Review-Reconstruction.md`
WAL-15	Review Remediation	`Part-03.0-Remediation-Reconstruction.md`
WAL-16	Documentation	`Part-03.0-Documentation.md`
WAL-17	Final Verification	`Part-03.0-Final-Verification.md`

4. Outcome

Part 03.0 established a durable multi-tenant data platform baseline. Part 02 added sessions/auth tokens (`382d8497298e`). Part 03.1 later hardened structural integrity and request-boundary security without replacing the entity map.

5. What Part 03.0 did not claim

- Production topology / trusted-proxy verification (Part 03.1 readiness matrix → Part 3.2)
- Composite cross-tenant foreign keys (added in Part 03.1)
- A contemporaneous independent-review finding register (reconstructed only; see WAL-14 report)

Evidence limitations (shared)

- Part 03.0 code primarily landed in git commit `838db64` (2026-07-09) under the commit message “chore: validate and stabilize backend foundation”, bundled with broader foundation scaffolding. There is no separate contemporaneous commit uniquely titled “Part 03.0”.
- Linear issues WAL-7…WAL-17 were created on 2026-07-18 as a historical execution hierarchy after the schema already existed in the repository.
- No contemporaneous Part-03.0 independent-review finding IDs, meeting notes, or signed review PDF exist in the repository (unlike Part 03.1’s report pack).
- Structural integrity remediations (composite cross-tenant FKs, soft-delete-safe unique indexes, missing `created_by`/`updated_by` FKs) are **Part 03.1** evidence (`f521d38`, revisions `3a2aec52f6d2` / `217e90d00cc0`) and are not attributed as silent Part 03.0 remediations.